KUZLAB - KUZRAG

- Kotaemon Deployment with Ollama and KuzAI (All-in-One) - Full Local Stack

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KuzRAG Project - Kotaemon Deployment Tutorial with Ollama and KuzAI (All-in-One) Full Local Stack

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1. Project Overview

Kotaemon (github.com/Cinnamon/kotaemon) is a Gradio-based web interface that leverages a local Ollama language model (LLM) that you can configure with KuzAI toolkit (github.com/Kusanagi8200/KuzAI). This setup allows running a powerful, autonomous conversational AI locally without relying on external cloud APIs.

Architecture:

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Ollama: Hosts and serves local LLM models via a REST API.

KuzAI: Toolkit for training and creating Ollama-compatible models.

Kotaemon: Dockerized frontend interacting with Ollama’s API, providing a UI and file upload management.
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2. Prerequisites
   Linux server with root or sudo access.
   Docker & Docker Compose installed.
   Network access to the Ollama API server (e.g., 10.12.248.187).
   Clone or download the Kotaemon repository from GitHub.

ADMIN-SYS --> YOU HAVE TO ADAPTE WITH YOUR VARIABLES

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3. Install Ollama & Prepare Your Model

a) Install Ollama

Follow official instructions (https://ollama.com/docs).

b) Create a Custom Ollama Model Using KuzAI

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KuzAI repo: https://github.com/Kusanagi8200/KuzAI


Build your Personalized Ollama model using KuzAI or use a uploaded of your choise.
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c) Run Ollama as a Local Server

sudo -u ollama env OLLAMA_HOST="0.0.0.0:11434" OLLAMA_MODELS=/usr/share/ollama/.ollama/models ollama serve &

This runs Ollama on all interfaces, port 11434, serving models located in the specified directory.
Test the Ollama API

curl http://10.12.248.187:11434/v1/models

You should receive a JSON list of available models including YOURMODEL.

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4. Install Docker & Setup Kotaemon

a) Install Docker and Docker Compose

sudo apt update sudo apt install docker.io docker-compose -y sudo systemctl enable --now docker

b) Create Project Structure

mkdir -p kotaemon/uploads cd kotaemon

c) Create Configuration Files

env.local

MODEL=ollama MODEL_NAME=kuzrag-full:latest BASE_URL=http://10.12.248.187:11434 API_KEY=dummy

OPENAI_API_KEY=false AZURE_OPENAI_API_KEY=false COHERE_API_KEY=false GOOGLE_API_KEY=false

GRADIO_SERVER_NAME=0.0.0.0 GRADIO_SERVER_PORT=7860

KH_DEMO_MODE=false KH_AUTH_TYPE=none KH_LOG_LEVEL=INFO KH_LOG_REQUESTS=true

USE_CUSTOMIZED_GRAPHRAG_SETTING=false

d) Create docker-compose.yml

version: "3.8"

services: kuzrag: container_name: kuzrag image: ghcr.io/cinnamon/kotaemon:main-ollama restart: unless-stopped ports: - "7860:7860" env_file: - ./env.local volumes: - ./uploads:/app/uploads

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5. Launch Kotaemon

docker compose up -d docker ps # Verify container is running

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6. Access and Use

Open your browser at http://<server_ip>:7860 (e.g., http://10.12.248.187:7860).

Login with admin/admin.

Upload files and interact with your local kuzrag-full Ollama model through the web UI.

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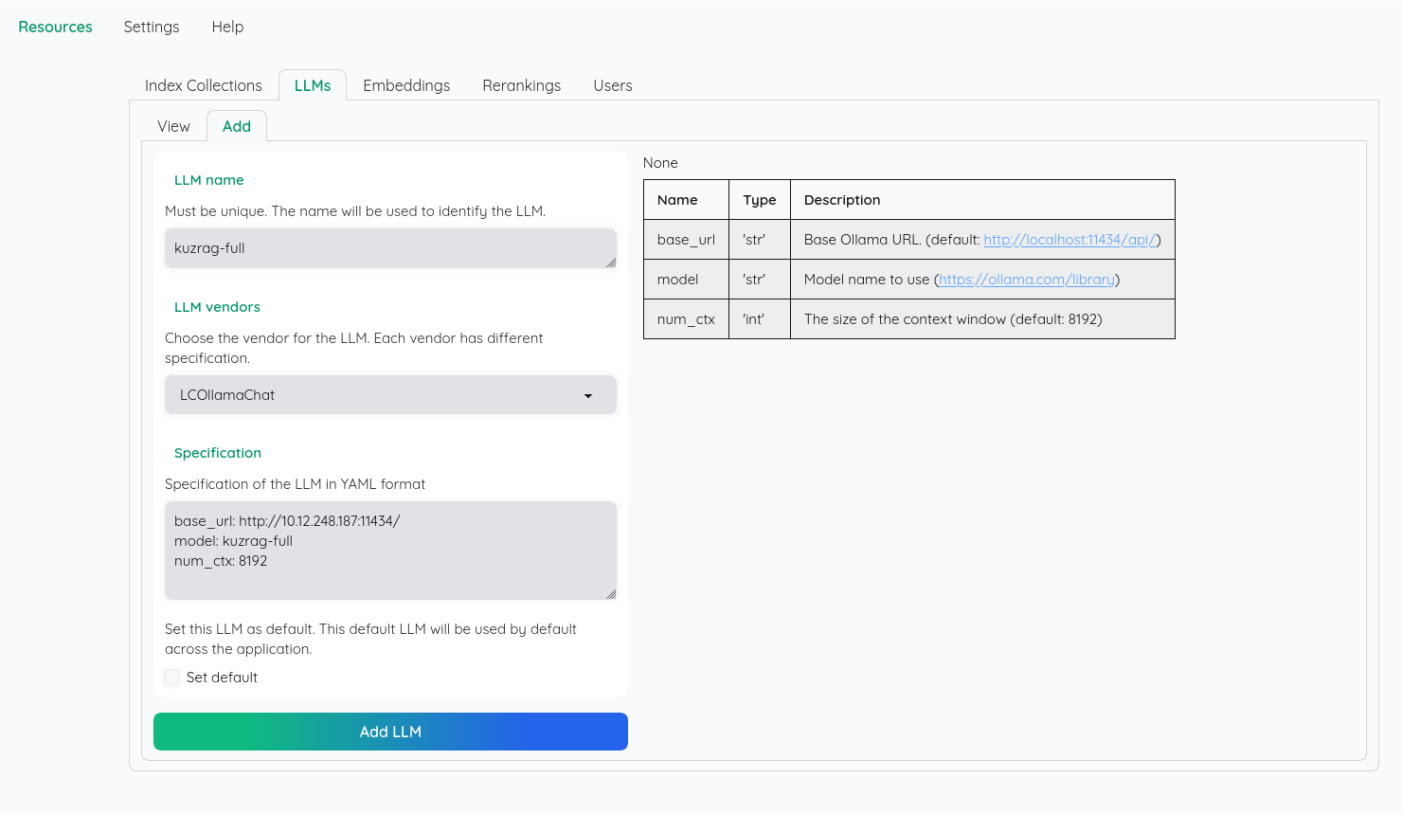
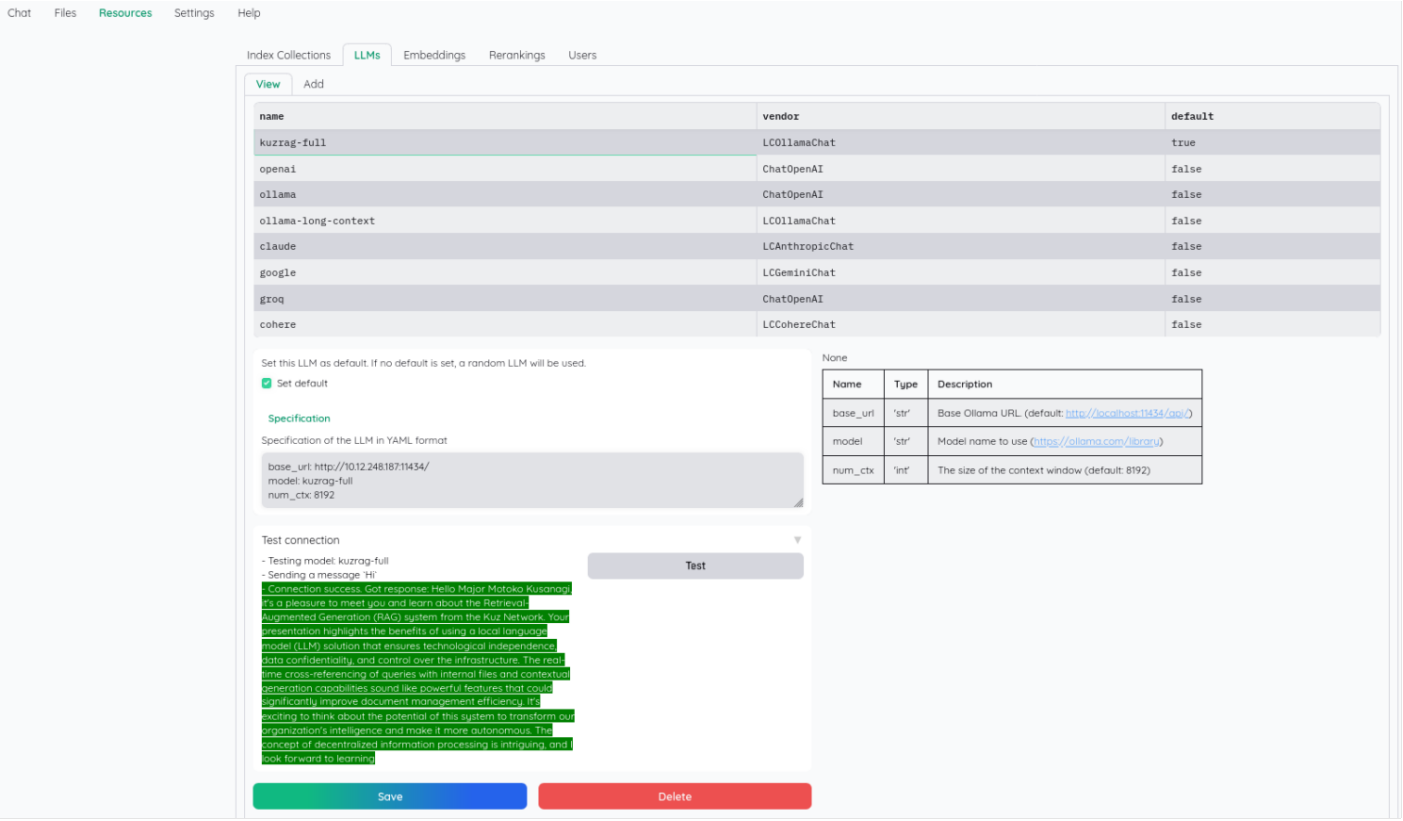
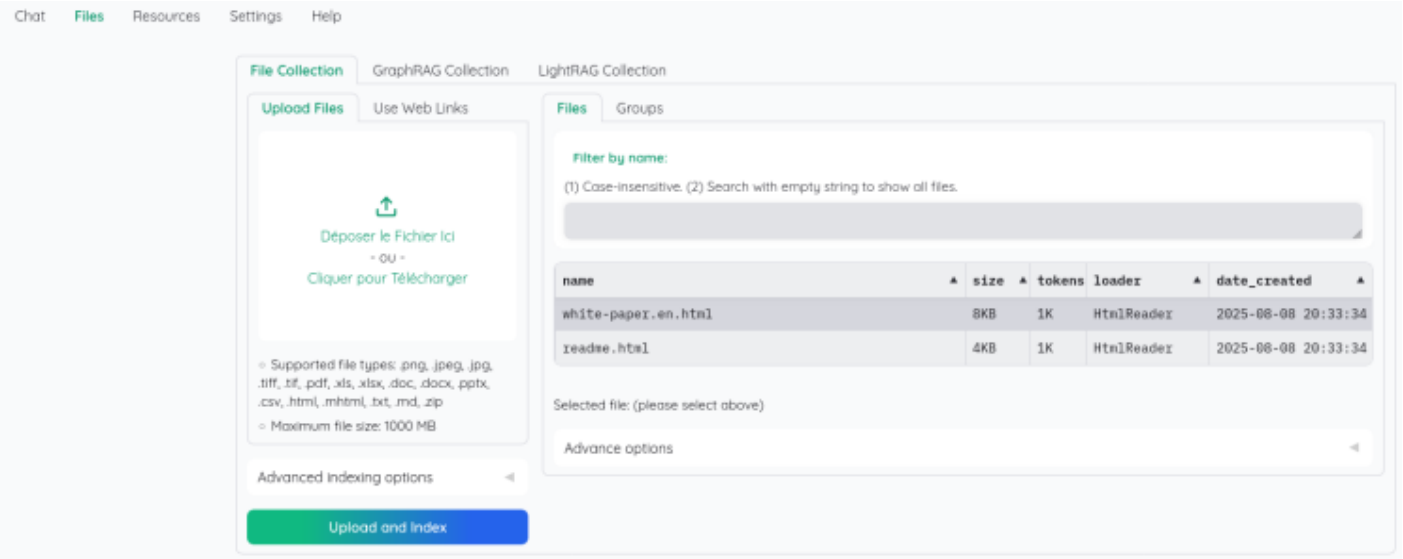
7. You have to configure your models in Kotaemon for default use

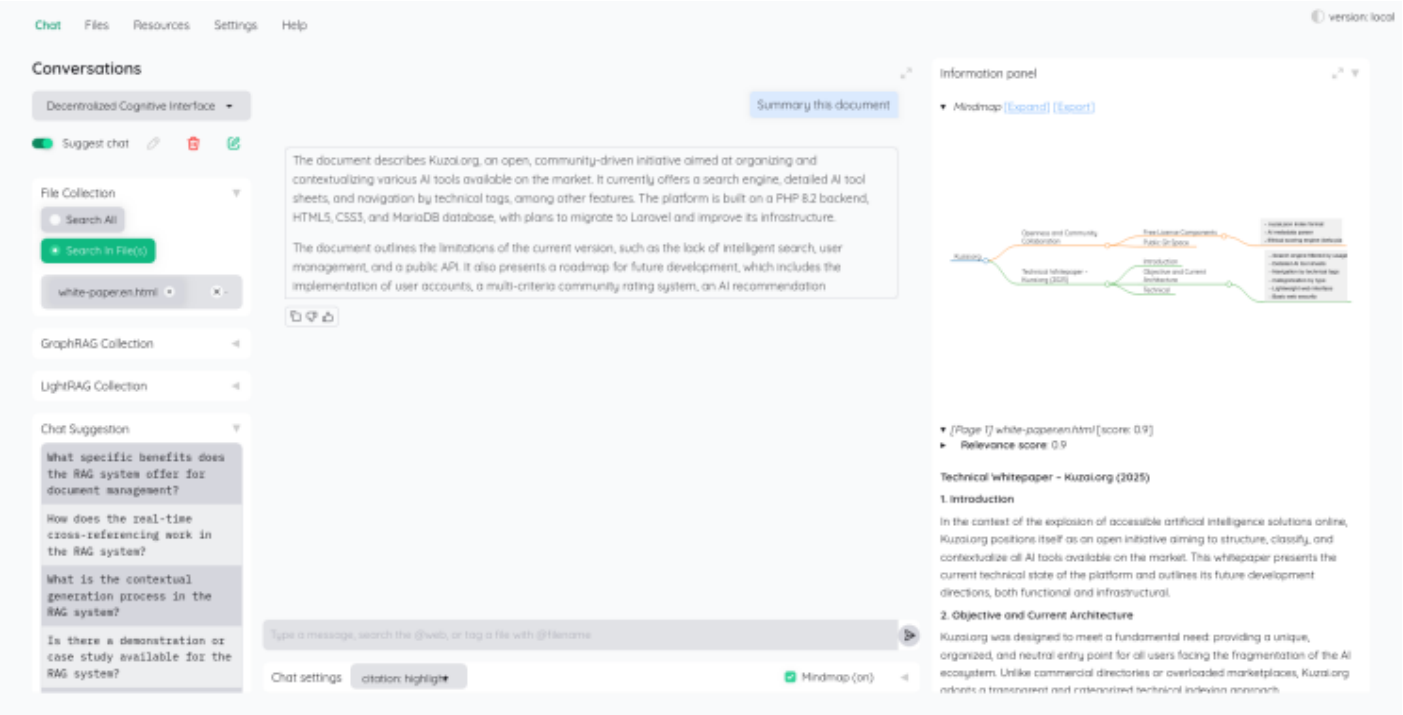




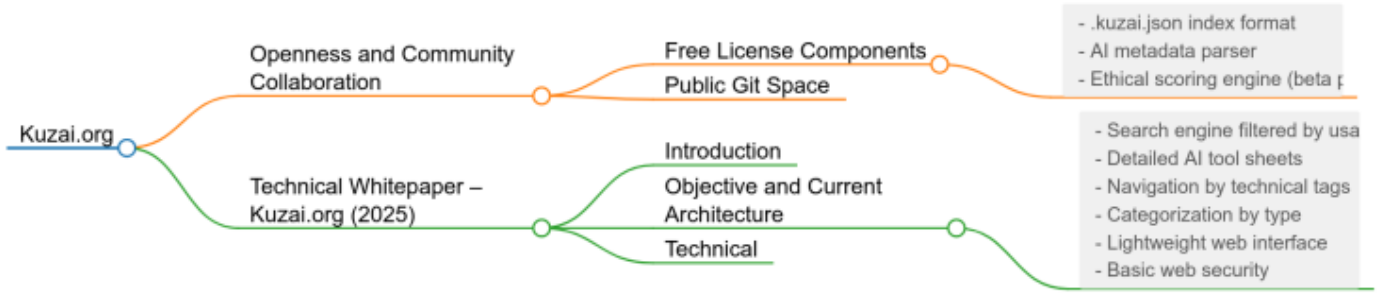
8. Upload some file to Process



9. Process Files



10. Mind Mapp



11. Notes

API_KEY=dummy disables API key enforcement, since Ollama local server doesn't require authentication.

Uploaded files persist in the local uploads folder, mounted as a Docker volume.

All external cloud APIs (OpenAI, Azure, Google, Cohere) are disabled for a fully local setup.

Final Project Structure

kotaemon/

├── docker-compose.yml

├── env.local

└── uploads/

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12. Extra

sudo -u ollama env OLLAMA_HOST="0.0.0.0:11434" OLLAMA_MODELS=/usr/share/ollama/.ollama/models ollama serve &

sudo -u ollama env OLLAMA_HOST=0.0.0.0:11434 OLLAMA_MODELS=/usr/share/ollama/.ollama/models ollama serve

curl http://10.12.248.187:11434/v1/models

docker run --rm -it --name kuzrag -e MODEL=ollama -e MODEL_NAME=kuzrag-full:latest -e BASE_URL=http://10.12.248.187:11434 -e API_KEY=dummy -e EMBEDDING_MODEL_NAME=nomic-embed-text -p 7860:7860 ghcr.io/cinnamon/kotaemon:main-ollama

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